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Scope of the Project

Our mission is to take United States socioeconomic data, public voting data, and census data to predict the potential votes for Democrats in the state of Georgia. More specifically, we would like to predict the percentage of voting for Democrats per county in Georgia with the number of registered voters in order to develop a campaign strategy for the Democrat or Republican party. Either party would use this information so that they may be able to strategically focus their campaign efforts in counties that would provide them the largest advantage. Time and campaign dollars are limited in any election and having a plan of action that would optimize their campaign would provide a significant advantage. We choose Georgia as our target population because there will be a runoff election in January 2021 that will determine which political party will seize the majority within the US Senate. Our process for achieving this goal is through testing four different statistical approaches to determine the model with the lowest mean squared error. Once we select the best model, we will assess our model with the counties within Georgia to rank the counties by the difference of expected votes minus what we observed in the 2020 Presidential election. The ranking of these counties will provide a clear picture of what counties the Democrats are performing well in and where more effort should be focused. A clear political strategy can then be created based on this ranking of counties and an optimization of how to spend campaign finances can be finalized. Lastly, the model has the potential to be repeatable to other states where either political party faces tough competition and close races. The usefulness of this model can change the course of elections and sway the political landscape of any state if used properly.

Cleaning and Merging datasets

All the training data was distributed into two data sets: 'socio-economic-data.csv' and 'president_county_candidate.csv'.Before training the respective models, it was necessary to clean and merge both datasets into one single table. This process was not trivial since both datasets were organized at different granular levels. While the socio-economic information was organized at city level, the presidential elections data was organized at county level. Moreover, the socio-economic data did not have a county column to allow a direct merge with the presidential elections data. Therefore, to merge both, it was necessary to use a third data-set, '*unzip.csv'*, which contained all the zip-codes for each city/county/state in the country.

In addition, it was essential to rearrange the socio-economic data by county, taking the mean of each county's corresponding features. Consequently, during this process it was required to drop the following features 'Predominant race', 'Population', and 'Total Population of State', since taking the mean of these variables did not make sense.

The final table had, beside all the socio-economic features the following new features:

- %DEM= percentage of democratic votes over the total voters
- %REP= percentage of republican votes over the total voters,
- County_name= name of each county
- Registered= total population who voted in the 2020 presidential elections.

Final selection of features

Our Model should be able to give a concrete estimate of votes per county based on the socio-economic characteristic of that particular county. Taking this into account, we will not include for our analysis the following features:

- County_name: The county name is unrelated with the result, moreover there are many repeated county names across the different states.
- %REP: Our Y variable could be expressed as %DEM and %REP, in this case the model will predict the %DEM, therefore %REP needs to be dropped.
- Registered: this feature was used to create the dependent variable we are predicting, therefore including this feature into the analysis will lead to a misunderstanding of its weight over the other covariates which constitutes pure socio-economic independent data. Since the goal of this analysis is to compare the potential number of votes given the socio-economic context of each state, this feature will not be included.

General Procedure to train and model selection

To select the best model to predict the percentage of Democrat votes, we use 10-fold cross validation. As previously mentioned, we use Georgia as test data, the rest of the data are training data. Then we implement the following procedures:

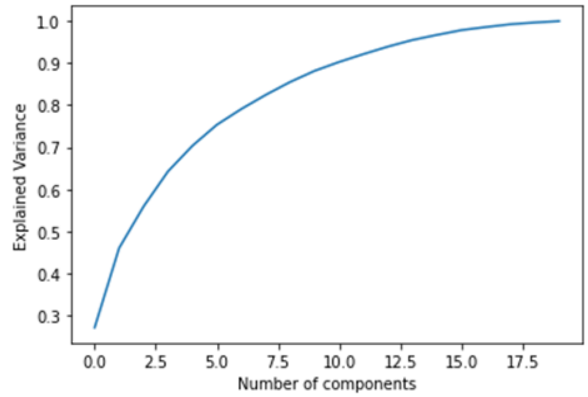
- In 10-fold Cross Validation, we divide the data set into 10 different parts
- We then remove the first part, fit the model on the remaining k-1 parts, and see how good the predictions are on the left out part by computing the MSE on the first part
- Then repeat these 10 different times taking out a different part each time
- By averaging the 10 different MSEs, we get an estimated validation MSE and select model with lowest MSE.

10-fold cross validation could address the issue of train-validation-test approach. The validation MSE of train-validation-test approach could be highly variable. Instead, the validation MSE of 10-fold cross validation is more stable.

Linear Regression

Linear Regression encompasses many forms, and we decided to implement a Normal Linear Regression and a Principal Component Regression (PCR). Under these broad regression types, we also included a Ridge and LASSO regression to understand if any one of these models best fit the dataset, with '%DEM' as our dependent variable and scaling the independent variables using Standard Scaler.

In the PCR analysis, we think that some of these independent variables are collinear, hence we decided to use a Principal Component Analysis that explains at least 95% of the variance.



The figure above shows that to explain at least 95% of the variance, we needed a minimum number of 14 components.

A summary of the different R-squared values and the MSE of 10-fold cross validation of the different linear regression is shown in this table.

Regression Type	R^2 value	MSE
Normal Linear Regression	0.51571	0.0123
Ridge Regression	0.51592	0.0123
LASSO Regression	0.51611	0.0123
PCR	0.48952	0.0129
PCR Ridge	0.48983	0.0129
PCR LASSO	0.48952	0.0129

According to the table, all the different regression models show marginal differences between each other, with R-squared values hovering around the 0.5 value. The best linear regression model is the Normal LASSO Regression with the lowest MSE value and the highest R-squared value.

Regression Tree

One method of assessment we analyzed was the use of a regression tree to predict the percentage of Democrat votes within a county based upon the socioeconomic information. We were able to successfully develop a tree that was interpretable to determine how much of a county would vote Democrat. The socioeconomic dataset consists of education, age, race, income, household size, and burden rates for homeowners and renters. The initial split based on owner burden is interesting because it is a clear break of areas with certain populations of homeowners. One reason for this separation could be between cities and suburban areas. In addition, the second largest split is for Graduate School education which could have something to do with areas of better education and more taxpayer dollars for schools. Our regression tree actually had a low MSE value which showed promising results. MSE = 0.01663235

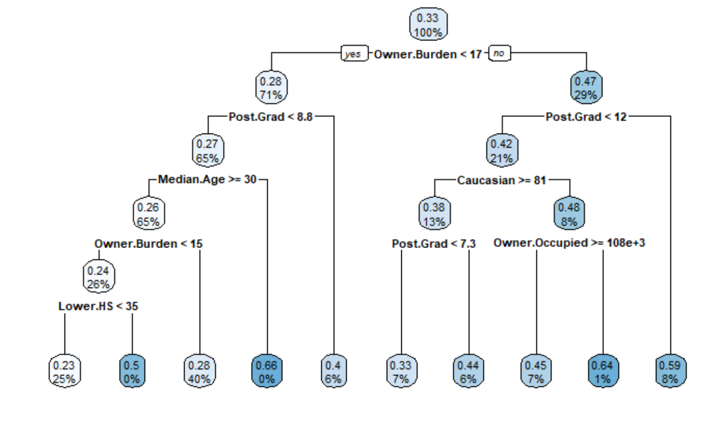


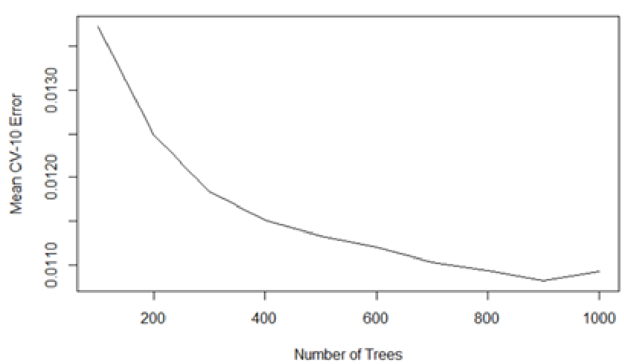
Figure. Regression Tree

Boosted Tree

This approach does not fit a single large decision tree to the data, instead the boosting approach learns slowly. Given the current model, we fit a decision tree to the residuals from the model. That is the new decision tree is fitted using the current residuals rather than the outcome Y.This model has three tuning parameters:

- Number of trees
- Interaction depth
- Learning rate (lambda).

Boosted trees will experience overfitting problems if the number of trees used to train the model is too large, therefore we need to select the best number of trees given a fixed learning rate and interaction depth. Since the growth of a particular tree takes into account the other trees that have already been grown, smaller trees are typically sufficient, and consequently we chose to fix the learning rate at the default value (0.001) and the interaction depth at 1. To select the best number of trees we used 10 folds cross validation.



In this case, we can observe that the best number of trees for the model will be 900. Once we have this parameter fixed the final boosted model is trained and tested using a Cross validation with 10 folds technique. The final mean error of the model is 0.0109542

Support Vector Regression

Support Vector Regression (SVR) gives us the flexibility to define how much error is acceptable in our model and will find an appropriate hyperplane to fit the data. Here, we use SVR to predict the percentage of Democrat votes. To avoid one feature dominating the function and making the estimator unable to learn from other features correctly as expected, we scaled the independent variables using Standard Scaler. This model has two tuning parameters:

- Epsilon: a margin of tolerance where no penalty is given to errors
- Cost (C): a regularization parameter. High C (cost) means the cost of error is increased.

The independent variable "% DEM" is from 0 to 1, so we set the epsilon between 0.01 and 0.2. Moreover, we set the range of C from 0.01 to 10. Then we use grid research with 10-fold cross validation to find the optimal parameters. The best parameters are C=0.6 and epsilon=0.02.

The MSE = 0.00811

Selection of Best Model

From these 4 models, we decided to select the model that best fits the training data by looking at the MSE scores, and choosing the one with the lowest score. As a result, our final model is the Support Vector Regression, which gives an MSE value of 0.00811 after performing 10-fold cross validation.

With this chosen model, we decided to test our findings on the state of Georgia, and figure out how accurate our predictions are when we compare it to the vote count for the 2 major parties in this recent 2020 election.

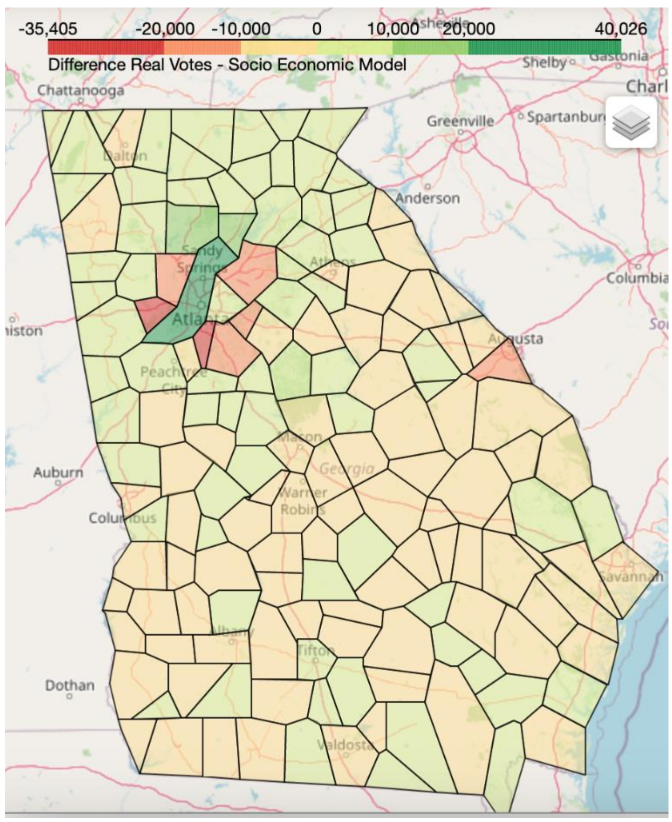
Our results show that the MSE of our test model is 0.0213. Using our LASSO linear regression as a supporting model, we estimated our coefficients to be summarized in this table below.

	Coefficients		
Registerd	0.010025	% college	-0.000000
% age 25-34	0.014206	Average HH size	-0.000000
% age 35-44	-0.005621	Median gross rent	-0.000000
% age 45-54	0.000000	Median value of an owner-occupied home	0.004370
% age 55-64	0.023988	Median HH income	-0.024309
% age 65 or older	-0.008080	Median owner cost burden	0.047982
Median age	-0.026834	Median renter cost burden	0.017071
% Bachelor's	0.020948	% Asian	0.003999
% HS diploma	-0.011309	% Black	0.009259
% less than HS diploma	0.007800	% Hispanic	-0.003819
% post grad	0.047880	% White	-0.000015

The table shows that % post grad and Median owner cost burden has the highest positive correlation in determining whether a voter will vote for a democratic party. This is supported by the Regression Tree method which separates the parties that voters vote for according to owner cost burden and postgraduate education. Owner cost burden is highly dependent on geographical location, and can indicate certain populations of homeowners. The higher the owner's cost burden, the higher the owner pays for his/her living expenses from income. It is also interesting to see that high education levels translate to more democratic voters. A reason could be that post graduates spend more time in liberal colleges and are thus, more liberal minded. The largest negative correlation comes from Median age and Median Household income. This means that the progressive democratic party appeals to younger voters, and those who are earning lower wages. From a political standpoint, this does make sense because the democratic party tends to support policies like improving the minimum wage and lowering taxes that appeals to workers who are earning below average income.

Final Results

Once the best model is selected, we proceed to estimate the percentage of votes the Democratic party should obtain based on the socio-economic context of each county. We then multiply this prediction by the total numbers of registered voters for each county obtaining the total number of Democrat votes by county. We now compare our model prediction with the real number of Democrats votes obtained by county, creating a heat map which allows us to visualize these results.



Applying these Results

The value added for the Democratic party is summarized through three major insights learned from the analysis of the socioeconomic data. First, we identified the features that have the highest weight for why Counties vote democrat. For example, counties with higher percentages of Post Graduate degrees vote democrat which could be because of their liberal educations and their free thinking mindset. Understanding these features can help the Democratic Party tailor campaign topics and advertisements towards the demographics that are within certain areas. The understanding of demographics should provide a higher turnout if the messaging to voters is effective and representative. Secondly, we were able to identify the counties that acted like Democrats and which did not. The simple ability to distinguish the counties that are similar to Democrats makes campaigning easier because you have voters that are already supportive to Democratic principles. With this knowledge, you will focus more on voter turnout rather than changing minds of undecided voters or Republicans. This will change the way the Democratic party spends campaign finances and their approach to increasing voter turnout. Lastly, we are also able to identify the counties the Democrats were not supposed to do well in but had a higher amount of votes than expected. The Democratic party could study these counties and their past approaches to campaigning there and hopefully identify the strategies they used. The insights gained from this analysis will provide the Democratic Party with strategies and areas of focus for their campaigning while increasing voter turnout and optimizing costs.

Next steps

In the future, we could use the total population who voted in the 2020 presidential elections as a dependent variable to run a linear regression to make inferences from the data. From the result, we could know what type of people are more likely to vote and what type of people don't prefer to vote. Then, we could focus on people who prefer not to vote, especially those who have a higher probability to vote for Democratic (for example, people with a graduate or professional degree). If we have more time, we would like to do more research on what strategy Democrats used in each county and figure out what is the most effective way to get the vote.

Appendix:

Definitions of Features in the Socioeconomic dataset

Age

- % age 25-34: Estimated percent of the population age 25 to 34 years, between 2014-2018.
- % age 35-44: Estimated percent of the population age 35 to 44 years, between 2014-2018.
- % age 45-54: Estimated percent of the population age 45 to 54 years, between 2014-2018.
- % age 55-64: Estimated percent of the population age 55 to 64 years, between 2014-2018.
- % age 65 or older: Estimated percent of the population age 65 years and older, between 2014-2018.
- Median age : Estimated median age of the population between 2013-2017.

Education

- % less than HS diploma: Estimated percent of population 25 years and older with less than a high school diploma or equivalent between 2013-2017.
- % HS Diploma: Estimated percent of population 25 years and older with a high school diploma or equivalent and no college between 2013-2017.
- % College: Estimated percent of the population 25 years and older whose educational attainment is some college, or an Associate's degree, between 2013-2017. Bachelor's
- % Bachelor’s: Estimated percent of population 25 years and older with a Bachelor's degree, between 2014-2018.
- % Post Grad: Estimated percent of population 25 years and older with a graduate or professional degree between 2013-2017.

Household and Spending

- Average household size: Estimated average household size, between 2014-2018. A household includes all the people who occupy a housing unit as their usual place of residence.
- Median gross rent: Estimated median gross rent for rental units with cash rent, between 2014-2018. Gross rent is the contract rent plus the estimated average monthly cost of utilities and fuels if these are paid by the renter). Gross rent is intended to eliminate differentials that result from varying practices with respect to the inclusion of utilities and fuels as part of the rental payment.
- Median value of an owner-occupied home: Estimated median value of an owner-occupied housing unit, between 2014-2018. The value is based on survey respondents' estimates of how much their properties and lots would sell for if they were for sale.
- Median household income: Estimated median household income in the past 12 months, as reported between 2014-2018. A household includes all the people who occupy a housing unit as their usual place of residence.
- Median renter cost burden: Median gross rent as a percentage of household income, between 2014-2018.
- Median owner cost burden: Estimated median selected monthly owner costs as a percentage of household income, for all owner-occupied housing units (with and without a mortgage), between 2014-2018.

Ethnicity

Data was obtained from the Census' American Community Survey 2013-2017 estimates

- % White: Estimated percent of the population that is White, by single classification of Census race, between 2013-2017.
- % Hispanic: Estimated percent of the population that is Hispanic or Latino between 2013-2017.
- % Black: Estimated percent of the population that is Black or African American, by single classification of Census race, between 2013-2017.
- % Asian: Estimated percent of the population that is Asian, by single classification of Census race, between 2013-2017.

Relevance of features according to Boosted tree model

var <chr>	rel.inf <dbl>
Median.owner.cost.burden	22.8452091
X..post.grad	20.2179803
Median.HH.income	7.2483848
Average.HH.size	4.7954635
Median.age	4.4750832
X..HS.diploma	4.0768531
Median.renter.cost.burden	4.0666997
X..Bachelor.s	3.8460015
X..age.25.34	3.5427496
X..age.55.64	3.5042786